

Predicting Sprint Performance Using Race Progression Data

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Abstract

Sprinting is a well-oiled mixture of reaction time, acceleration, max velocity, and speed maintenance through the finish line. While final times are commonly used to judge performance, the race metrics before are just as valuable in the information that they provide of how the athlete did it. This research investigates the question: How accurately can machine learning models predict sprint performance using race progression data across the 100m, 200m, and 400m events? In order to answer this question, publicly available race records were preprocessed to create three datasets containing reaction time, wind, and time and velocity splits for the 100m, 200m, and 400m races. Four supervised regression models—Linear Regression, Random Forest Regression, Gradient Boosting Regression, and HistGradientBoosting Regression—were trained on the data to be able to predict final times from early and intermediate race measurements. The datasets contained 2,631 samples for the 100m event, 2,345 samples for the 200m event, and 2,884 samples for the 400m event. Model performances were evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 score. HistGradientBoosting Regression earned the best results for the 100m, Gradient Boosting Regression rose to the top for the 200m, and Random Forest Regression finished #1 in accuracy for the 400m. All these results explain how machine learning can use race progression data for prediction models, and that the ideal model varies based on the characteristics of the sprint event being analyzed.

1 Introduction

Sprinting events in track and field like the 100-meter, 200-meter, and 400-meter races all need athletes to push themselves in terms of acceleration, reach maximum velocity, and hold that top speed through the finish line. Fractions of a second can make all the difference. Performance has been traditionally judged by the final times of an athlete in each of the races. Although, that final time tells us nothing on how an athlete achieved success or failure, losing the knowledge of which parts of the race were most influential to their performance.

Race progression data, such as the time and velocity splits of an athlete through a race in its entirety, gives a fuller picture on how an athlete did. For example, an athlete in the 100m who attains max velocity faster has a different profile to the sprinting of an athlete whose acceleration is more progressive but can maintain speed more efficiently. Yet, their ending times being the same make the athletes seem identical. Understanding their differences can help both sprinters and their coaches find flaws in their performance and develop more productive training regimens.

This project investigates the research question: How accurately can machine learning models predict sprint performance using race progression data across the 100m, 200m, and 400m events?

The end goal of this research project is to create predictive models capable of accurately estimating final sprint times based on early and intermediate metrics. By analyzing which features impact performance the most, this research also intends to show how the phases of a race influence its final performance.

This project is set up as a supervised machine learning regression task. The input data are the velocity and time splits for each race: every 10m for the 100m and every 50m for the 200m and 400m. The output from each model is a continuous numerical value: the athlete's final race time. Four regression algorithms were used to establish which was most effective in predicting final sprint performance.

This research is personally meaningful to me as I run all three events for Saratoga High School. In my own races, I am often behind everyone else in the first half, and only gain ground and reach a podium spot in the second half. Through my own experiences, I grew interested in understanding acceleration and race progression. By combining machine learning with the sprint data of the world's best athletes, this project is the union of my inquisitiveness in the sport of track and field with data science while diving deep into methods of athletic analysis.

2 Background

Previous research has shown that the different phases in a race aren't all equal in their contributions to the final time. Rather, it varies between events. The 100-meter dash is heavily dependent on an athlete's explosiveness out of the blocks and ability to hit max velocity as soon as possible, whereas the 200-meter and 400-meter rely more on athletes maximizing speed over a longer distance while fighting the fatigue.

Tønnessen, Haugen, and Shalfawi (2013) [1] examined the reaction time of the top athletes at the World Championships and found that it was associated with 100m performance. Their study analyzed 1,319 elite sprinters and confirmed that small reaction time differences in the start phase of races influenced final times. Even though their work mainly focused on reaction time, it speaks to the value that evaluating early-race metrics has in predicting ending performances.

Maćkała, Fostiak, and Kowalski (2015) [2] investigated determinants of acceleration in the 100m dash and specified the bonds between acceleration, motor ability, anthropometric traits, and the final times that they led to. Their results show that the early phase of sprinting events has an impact on a sprinter's overall performance.

In more recent times, machine learning was implemented to the prediction of sprints. Tam and Yao (2024) [3] used these methods to model the velocity curve of the 100m and found a strong correlation between maximum velocity and final time. Their findings established the fact that models can depict the complex connections between the increase in a sprinter's velocity to their ending time. Unlike normal paths to reach the same goal, machine learning can recognize nonlinear patterns in the data that aren't as obvious when using simple correlations.

Even though previous studies have delved into race metrics like reaction time, velocity, and biomechanics, this research broadens upon all others by applying it to all sprinting events. As opposed to solely relying on the 100m, this project goes into the 100m, 200m

and 400m races to determine how the factors between each event affect the predictions of their final times.

3 Dataset

For sprint performance prediction data, three datasets consisting of the 100m, 200m, and 400m were used. The data contains measurements that play a big role in all the races, made up of the wind at that time, the reaction time, and the time split with velocity at specific distances corresponding to which event is being analyzed. The intent with using the holistic data instead of only final times is to allow the machine learning models to see how the athlete developed, maintained, and controlled their speed from start to finish.

The initial dataset used was from a publicly available GitHub repository housing sprint data ready for machine learning use. This dataset was first selected as it had the necessary race variables in an organized manner for the models to use as inputs. In order to increase the amount of data and the variety of sprinters employed, the data was expanded to the elite race logs of AthleteFirst [4].

Because AthleteFirst’s publicly available race databases store information in a PDF format rather than in a CSV, preprocessing was required before it could be applied to the models. The race records were first extracted from the PDFs using MinerU, transforming them into markdown files. The markdown files were then processed using Python code generated with Gemini assistance to convert the obtained data into usable CSV files fit for machine learning.

The final datasets comprised of:

Table 1: Dataset features across the 100m, 200m, and 400m events.

Event	# of Samples	# of Features	Feature Type
100 Meter	2631	26	Wind speed, reaction time, 10-meter split times and velocity measurements
200 Meter	2345	15	Wind speed, reaction time, 50-meter split times and velocity measurements
400 Meter	2884	22	Wind speed, reaction time, 50-meter split times and velocity measurements

The 100M dataset included measurements at 10m intervals. This was done to display the acceleration and max velocity of an athlete during the race. For the 200m and 400m, datasets included measurements at 50m intervals, differing from the 100m in that the models studied the acceleration and speed maintenance needed for the longer races.

The target output was the athlete’s ending times. Therefore, the models used the race information as input features to identify the relationship between starting and intermediate splits against final performance, then attempting to predict it itself.

The datasets were divided into training and testing subsets with a `random_state = 42` for precise results every time. The exact training and testing split is 80% and 20%.

4 Exploratory Data Analysis

4.1 100M Event Visualizations

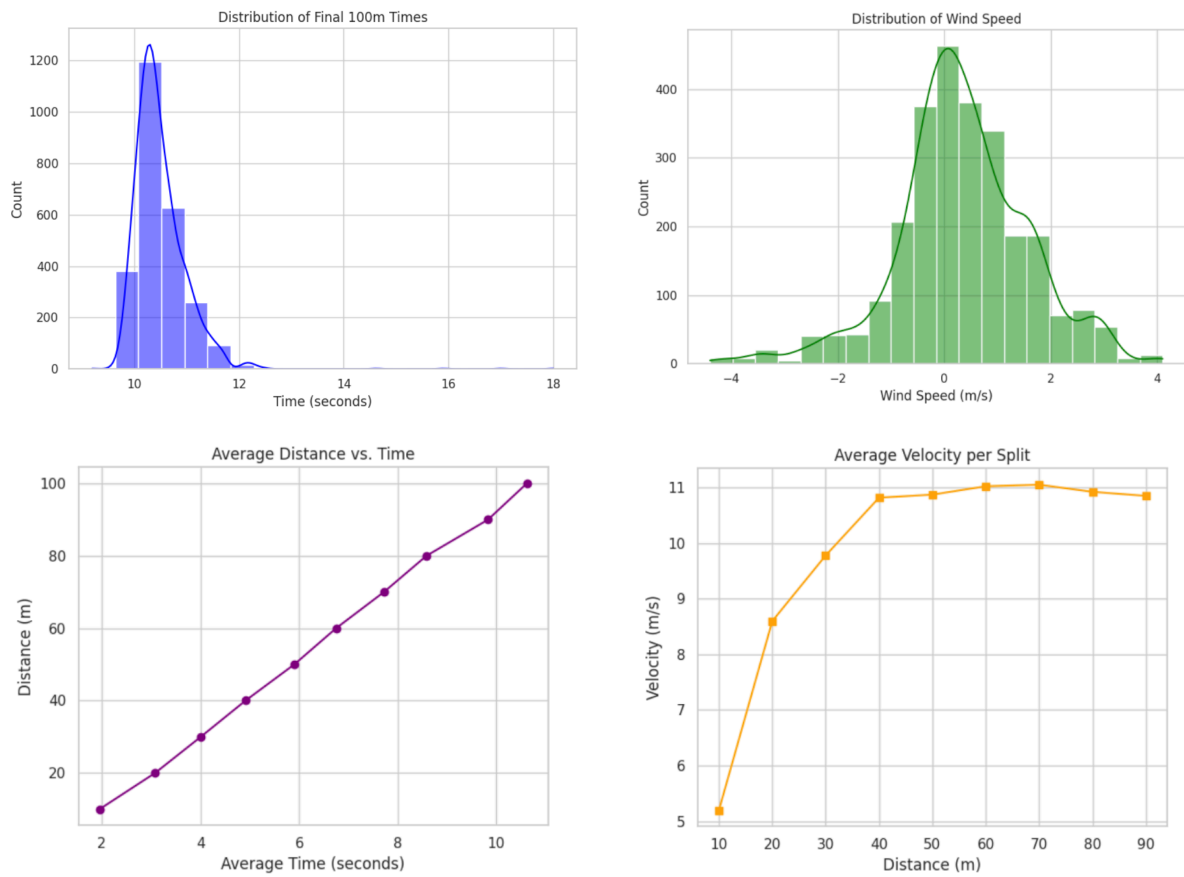


Figure 1: Data visualizations for the 100M event datasets.

4.2 200M Event Visualizations

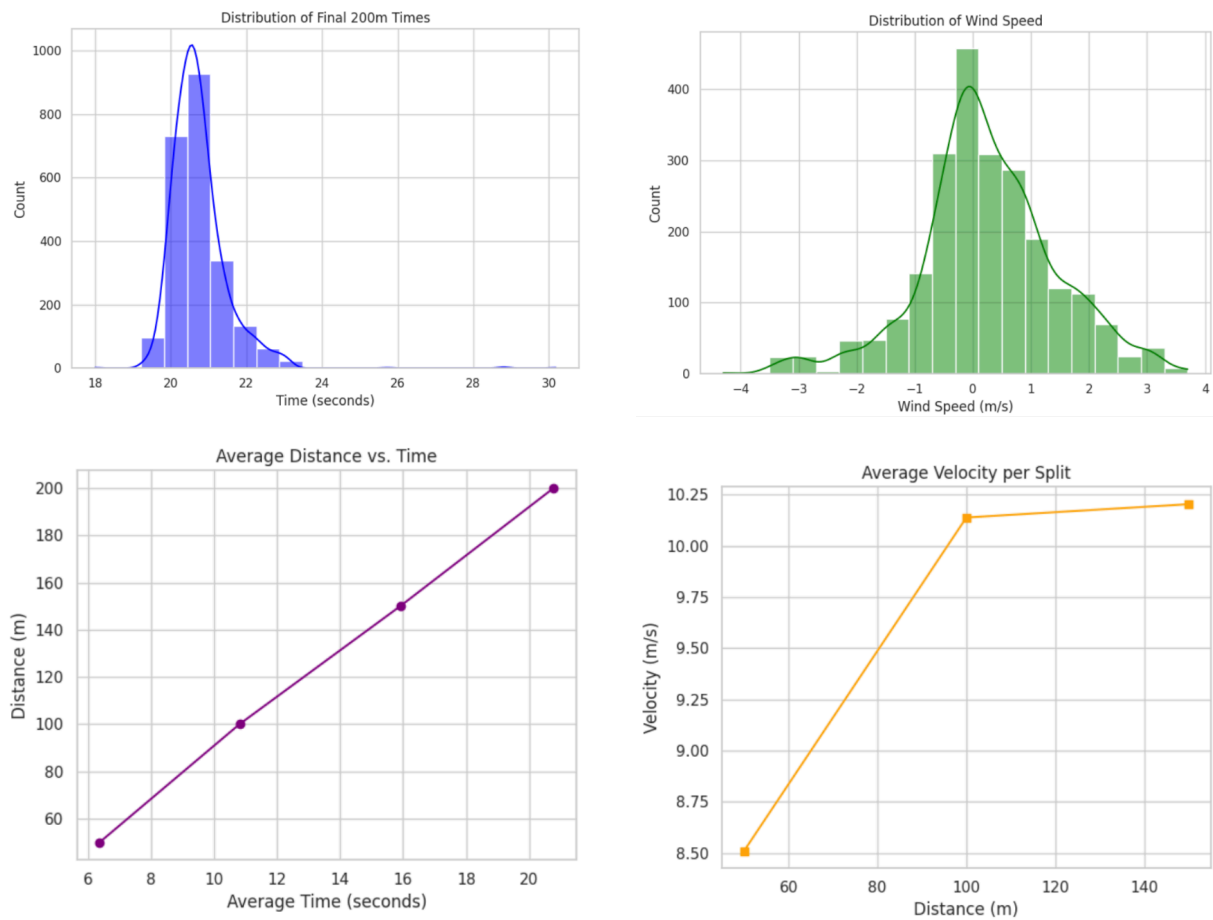


Figure 2: Data visualizations for the 200M event datasets.

4.3 400M Event Visualizations

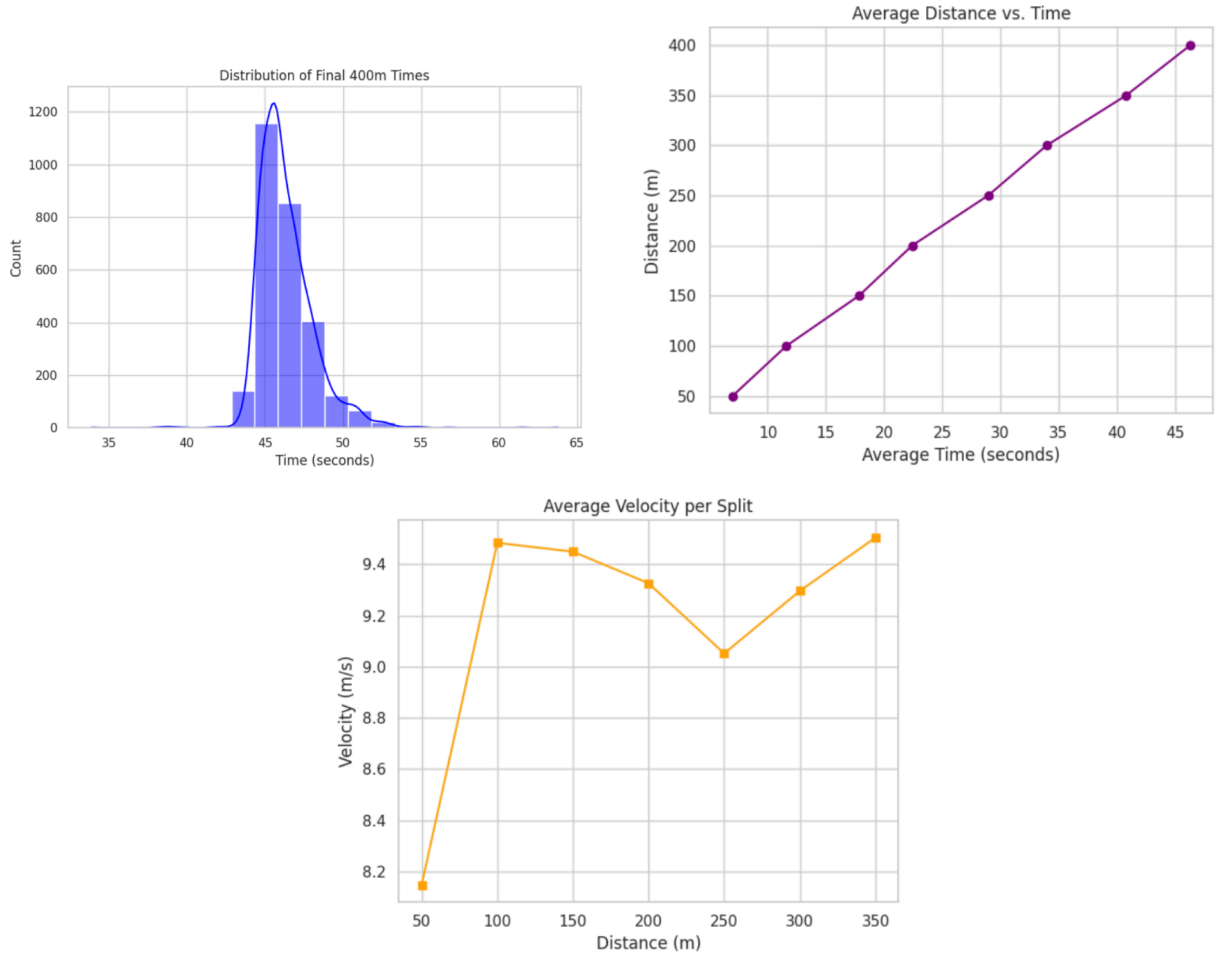


Figure 3: Data visualizations for the 400M event datasets.

5 Methodology / Models

This research used supervised machine learning for predicting final times from the race progression datasets. Because the output is a measurement, not a category, regression models were used. Four different models were applied and contrasted: Linear Regression, Random Forest Regression, Gradient Boosting Regression, and HistGradientBoosting Regression.

Linear Regression was used as a baseline model since it's a simple method for measuring the relationship between race split input values and final time output values. The model assumes that the association between these 2 variables can be portrayed through the linear equation:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n \quad (1)$$

Where y is the race time predicted by the model, x is the split times and velocities given to the model, and β is the model's learned coefficients. Even though sprinting is much more than a linear relationship, as it is more often affected by non-linear factors, using a Linear Regression gives a useful benchmark for seeing how much farther more complex models reach.

Random Forest Regression combines the predictions of many decision trees. Each tree

is trained on a bootstrap sample of the training data and considers a random subset of features at each split, so the trees make partly independent errors. Averaging their predictions reduces overfitting and improves stability.

Gradient Boosting Regression builds a sequence of weak models, each fitted to the residual errors of the model before it. Each new model therefore focuses on the cases its predecessors handled worst.

HistGradientBoosting Regression was included for its efficiency on larger datasets. Instead of evaluating every unique value of a continuous feature when searching for split points, it first sorts feature values into a fixed number of discrete bins, which substantially reduces the number of candidate splits that must be tested.

The general workflow was:

1. Load and preprocess the sprint progression datasets.
2. Separate race progression features from the final times.
3. Split the data into 80% training and 20% testing subsets.
4. Train each regression model using the training data.
5. Generate predictions on unseen testing data.
6. Compare model performance using the evaluation metrics.

The performance of the models were judged based on Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R^2 score.

6 Results and Discussion

The end goal of the research was to figure out how accurately machine learning models can predict sprint performance using race progression data across the 100m, 200m, and 400m events.

All in all, the results of the models substantiate the fact that machine learning models are able to predict final sprint times using prior sprint data. The strongest models had high R^2 scores, highlighting how velocity and time splits have the necessary information to foresee an athlete's performance. However, no model outperformed the other approaches universally. Instead, the best model varied based on the distance being investigated.

100M Results: The 100M dataset had the strongest predictive performance for all 3 of sprinting events. The HistGradientBoosting Regression model performed best with an MSE of 0.0108, RMSE of 0.1040, and R^2 score of 0.9379.

Table 2: Model performances for the 100M Event.

Model	MSE	RMSE	R^2
Linear Regression	0.0530	0.2302	0.6958
Random Forest Regression	0.0157	0.1255	0.9096
Gradient Boosting Regression	0.0291	0.1706	0.8329
HistGradientBoosting Regression	0.0108	0.1040	0.9379

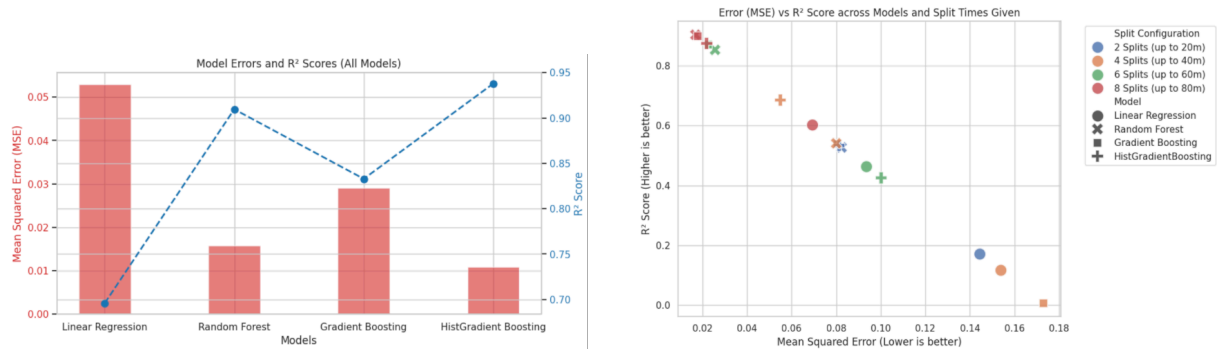


Figure 4: Prediction errors and R^2 scores across 100m models.

200M Results: The 200m data had a lower accuracy of prediction when compared to the 100m and 400m datasets. The model with the best performance was Gradient Boosting Regression, with an MSE score of 0.1738, RMSE of 0.4170, and R^2 score of 0.7946.

Table 3: Model performances for the 200M Event.

Model	MSE	RMSE	R^2
Linear Regression	0.3018	0.5494	0.6435
Random Forest Regression	0.2325	0.4822	0.7253
Gradient Boosting Regression	0.1738	0.4170	0.7946
HistGradientBoosting Regression	0.2167	0.4655	0.7441

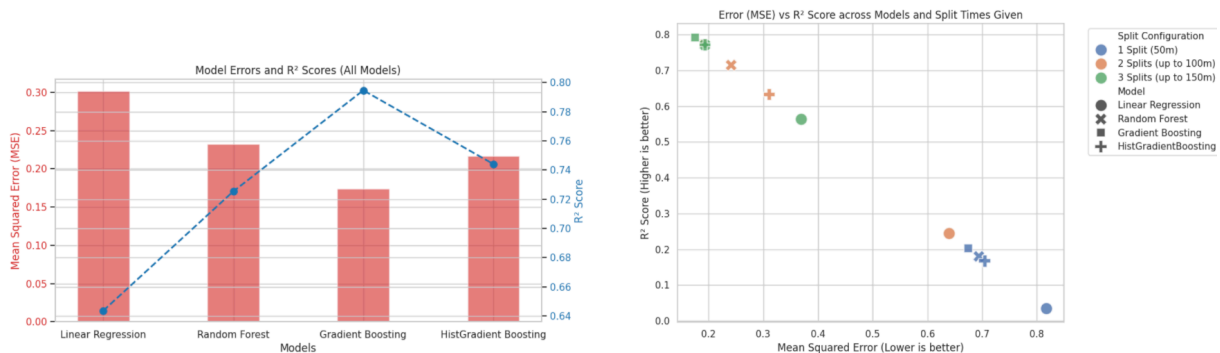


Figure 5: Prediction errors and R^2 scores across 200m models.

400M Results: The 400m dataset had a solid predictive performance. Random Forest Regression was the best model, illustrating the complex interactions of different race phases in the longer sprinting event.

Table 4: Model performances for the 400M Event.

Model	MSE	RMSE	R^2
Linear Regression	1.6546	1.2863	0.4502
Random Forest Regression	0.2469	0.4969	0.9180
Gradient Boosting Regression	0.3027	0.5502	0.8994
HistGradientBoosting Regression	0.3091	0.5560	0.8973

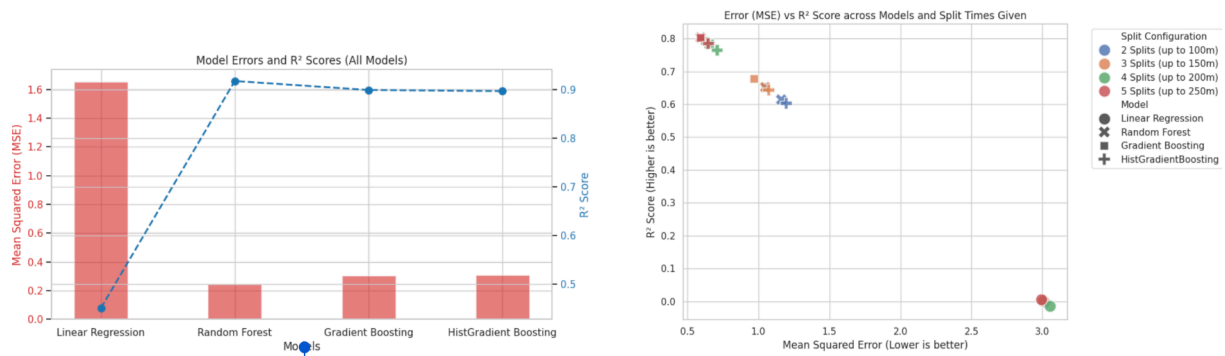


Figure 6: Prediction errors and R^2 scores across 400m models.

Across all 3 of the sprint distance events, the results exhibit how different machine learning models are more appropriate for different sprint predictions. HistGradientBoosting Regression was best for the 100m dash, where quick acceleration and high velocity are the name of the game. Gradient Boosting Regression produced the strongest results for the 200m, since athletes have to balance both max velocity and speed maintenance. Random Forest Regression ended as the best for the 400m event, due to its longer race introducing more muscle fatigue and different pacing strategies.

7 Conclusion

This research project explored if machine learning models could accurately predict sprint performance using race progression data from the 100m, 200m, and 400m events. The results proved that machine learning models can apply both time and velocity splits to predicting final times. HistGradientBoosting Regression was number one for 100m times, Gradient Boosting Regression executed the best for 200m, and Random Forest Regression was the peak for 400m.

These conclusions denote that different sprint distances have various effects on the models used for their predictions. The variations between races underline the significance of taking into account the structure of races and the performance they demand when constructing prediction models. Future improvements could include incorporating more variables such as biomechanical measurements when sprinters run, or the sprinters' height and weight.

References

- [1] Tønnessen, E., Haugen, T., & Shalfawi, S. A. I. (2013). Reaction Time Aspects of Elite Sprinters in Athletic World Championships. *Journal of Strength and Conditioning Research*, 27(4), 885–892. <https://doi.org/10.1519/JSC.0b013e31826520c3>
- [2] Maćkała, K., Fostiak, M., & Kowalski, K. (2015). Selected Determinants of Acceleration in the 100m Sprint. *Journal of Human Kinetics*, 45(1), 135–148. <https://doi.org/10.1515/hukin-2015-0014>
- [3] Tam, C. K., & Yao, Z.-F. (2024). Advancing 100m Sprint Performance Prediction: A Machine Learning Approach to Velocity Curve Modeling and Performance Correlation. *PLOS ONE*, 19(5), e0303366. <https://doi.org/10.1371/journal.pone.0303366>
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